

25PY101: Engineering Physics

Dr. Sreekar Guddeti

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Summative preparation

QM, QBT

Instructions

- Answer all questions.
- Show all the calculation steps clearly.
- Draw neat and properly labeled diagrams.
- Use SI units wherever necessary.

Questions

Q1. Photo-electric effect

The work function ϕ of Au is 4.90 eV.

- (a) A blue photon is incident on Au. Does it emit photo-electron? If yes, what is the stopping potential required to have zero photo-current.
- (b) Determine the threshold frequency ν_0 for photon emission of electrons for Au.

Repeat the above steps for Cs with work function 1.90 eV.

Q2. Graphical / Sketch-Based Question

Draw Fermi level in energy band diagram for the following under thermal equilibrium

1. p-type semiconductor,
2. n-type, and
3. p-n junction diode.

Answer

Aim:

Topic:

Given Quantities:

Working Formulas:

Steps: